

SPRINT 2 · THE GHOSTWRITER

The Draft Machine

Replies That Don't Sound Robotic

MS2.2 · Wednesday

"Thread in, draft out — in your voice, every time."

-30 min · Groq API · One-ask rule · Tone matching · Length control

Yesterday — Context Is King (MS2.1)

What we built:

- ✓ `tone_profile.json` — your writing style
- ✓ `past_replies.json` — 2-3 real emails you sent
- ✓ `context_builder.py` — assembles the full prompt

Result: a perfectly structured prompt with
your tone + thread history + examples

The prompt structure:

SYSTEM PROMPT

- Your persona + tone rules
- Past reply examples (few-shot)

USER PROMPT

- Full thread history
- "Draft a reply"

Today: we SEND this to an LLM
and get a real draft back.

Today's Goal

Thread → Context Builder → Groq LLM → Draft Reply

- Build `draft_machine.py` — the function that actually calls the LLM
- Use Groq's free tier (llama-3.3-70b-versatile model)
- Apply 3 key rules to make replies not suck:
 1. The One-Ask Rule (one clear question/response per email)
 2. Tone Matching (sound like YOU, not a chatbot)
 3. Length Control (short thread = short reply)
- Output: a `draft_reply()` function you'll use for the rest of the sprint

Rule #1: The One-Ask Rule

Every email should have **ONE** clear question or **ONE** clear response.

BAD (multiple buried asks):

Hi,
Can you review the budget? Also, when is the meeting? And did you talk to Sarah? Let me know about the vendor too. Oh and can you update the doc?
Thanks

GOOD (one clear ask):

Reviewed the budget – looks good overall.
One flag: marketing is 20% over last Q.

Should I dig into that or move forward as-is?

Best, Rahul

We encode this directly into the system prompt as a hard rule.

Rule #2: Tone Matching

The draft should sound like YOU wrote it — not ChatGPT.

Generic AI reply:

Dear Team,

I hope this email finds you well. I wanted to follow up regarding the Q3 budget document. After careful consideration, I have reviewed the marketing allocation and would like to share my thoughts at your earliest convenience.

Please don't hesitate to reach out.
Best regards

Tone-matched reply (sounds like Rahul):

Reviewed the budget doc. Here's my take:

1. Overall numbers look solid
2. Marketing paid ads – agree we're overspending. Cut 20% and shift to organic?

Want me to draft the reallocation or just flag it in the doc?

Thanks, R

How: few-shot examples + explicit persona rules in system prompt

Rule #3: Length Control

AI loves to ramble. We force brevity.

- Match thread energy: short question → short reply
- Quick replies: 2–3 sentences max
- Substantive replies: 5 sentences max, use numbered points
- NEVER write more than the sender wrote (unless essential)

How we enforce this:

1. System prompt: explicit length rules
2. max_tokens=500: hard cap on output length
3. Few-shot examples: all examples are short → model mimics

Code Structure: draft_machine.py

```
from groq import Groq
from context_builder import assemble_context

def draft_reply(thread):
    # 1. Assemble context (tone + examples + thread)
    context = assemble_context(thread)

    # 2. Add drafting rules (one-ask, length, no-filler)
    system = context['system'] + DRAFTING_RULES

    # 3. Call Groq
    client = Groq(api_key=os.environ['GROQ_API_KEY'])
    response = client.chat.completions.create(
        model='llama-3.3-70b-versatile',
        messages=[
            {'role': 'system', 'content': system},
            {'role': 'user', 'content': context['user']}
        ],
        temperature=0.7,
        max_tokens=500
    )

    # 4. Return the draft
    return response.choices[0].message.content
```

The Drafting Rules (added to system prompt)

CRITICAL DRAFTING RULES:

1. ONE-ASK RULE:

Every email has exactly ONE clear question or ONE clear response. Never bury multiple asks.

2. LENGTH CONTROL:

- Match thread energy (short Q = short reply)
- Quick replies: 2-3 sentences max
- Substantive: 5 sentences max, use numbered points
- NEVER write more than the sender wrote

3. NO AI FILLER:

- Never say 'I hope this finds you well'
- Never say 'Thank you for reaching out'
- Never start with 'I' – start with content

4. STRUCTURE:

Acknowledge briefly -> Response -> ONE next step

Let's Build It!

Thread → Context Builder → Groq → Draft Reply

Free tier: Groq (llama-3.3-70b-versatile) • No credit card needed

Demo: Before vs After

BEFORE (generic AI):

Dear Colleague,

I hope this email finds you well. Thank you for your message regarding the Q3 budget document. I have taken the time to thoroughly review the marketing allocation as requested.

After careful analysis, I would like to share that I concur with your assessment regarding the paid advertising spend. I believe we could optimize our budget allocation in this area.

Please let me know how you would like to proceed. I am happy to discuss further at your earliest convenience.

Best regards,
Rahul

AFTER (our Draft Machine):

Reviewed the budget doc. My take:

1. Overall numbers look solid
2. Paid ads – agree, we're overspending.
~20% cut seems safe without losing reach if we shift to organic.

Want me to draft the reallocation or just flag it in the doc for Thursday's discussion?

Thanks, R

Same thread. Same person. Wildly different quality.

Why Groq? (Free Tier)

- Free tier: 30 requests/minute, no credit card required
- llama-3.3-70b-versatile: excellent at following instructions
- Fast inference: replies in < 2 seconds
- Good enough for email drafts (you review before sending anyway)

Setup: console.groq.com/keys → create key → export GROQ_API_KEY

Paid alternatives (for production):

- Claude API (Anthropic) — best tone matching
- OpenAI GPT-4o — solid all-rounder

Paid Alternative: Claude API

```
# Same structure – just swap the client
from anthropic import Anthropic

def draft_reply_claude(thread):
    context = assemble_context(thread)
    system = context['system'] + DRAFTING_RULES

    client = Anthropic() # uses ANTHROPIC_API_KEY env
    response = client.messages.create(
        model='claude-sonnet-4-20250514',
        max_tokens=500,
        system=system,
        messages=[
            {'role': 'user', 'content': context['user']}
        ]
    )
    return response.content[0].text

# Claude is better at tone matching and nuance.
# Cost: ~$0.003 per email draft (Sonnet).
# For this course: Groq free tier is plenty.
```

Today's Output

1. `draft_machine.py` with:
 - `draft_reply(thread)` — main function
 - One-ask rule baked into system prompt
 - Tone matching via `context_builder`
 - Length control via prompt rules + `max_tokens`
2. Working pipeline:
Thread → `context_builder` → Groq API → Draft
3. A draft that sounds like YOU wrote it
(not a chatbot, not corporate filler)

Recap + Assignment

Today we built:

- ✓ `draft_reply()` function calling Groq
- ✓ One-ask rule in system prompt
- ✓ Tone matching with examples
- ✓ Length control (prompt + `max_tokens`)
- ✓ Before/after comparison — night and day

Your assignment:

- Get a Groq API key (free, 30 sec)
- Run `draft_machine.py` on the sample thread
- Try it on a REAL thread from your inbox
- Tweak the drafting rules if needed
- Share your best draft in Slack

Tomorrow: "Human in the Loop" — approve, edit, or reject before sending.

See You Tomorrow!

Thursday: "Human in the Loop" — the approve/edit/reject gate

#MyAIChiefOfStaff